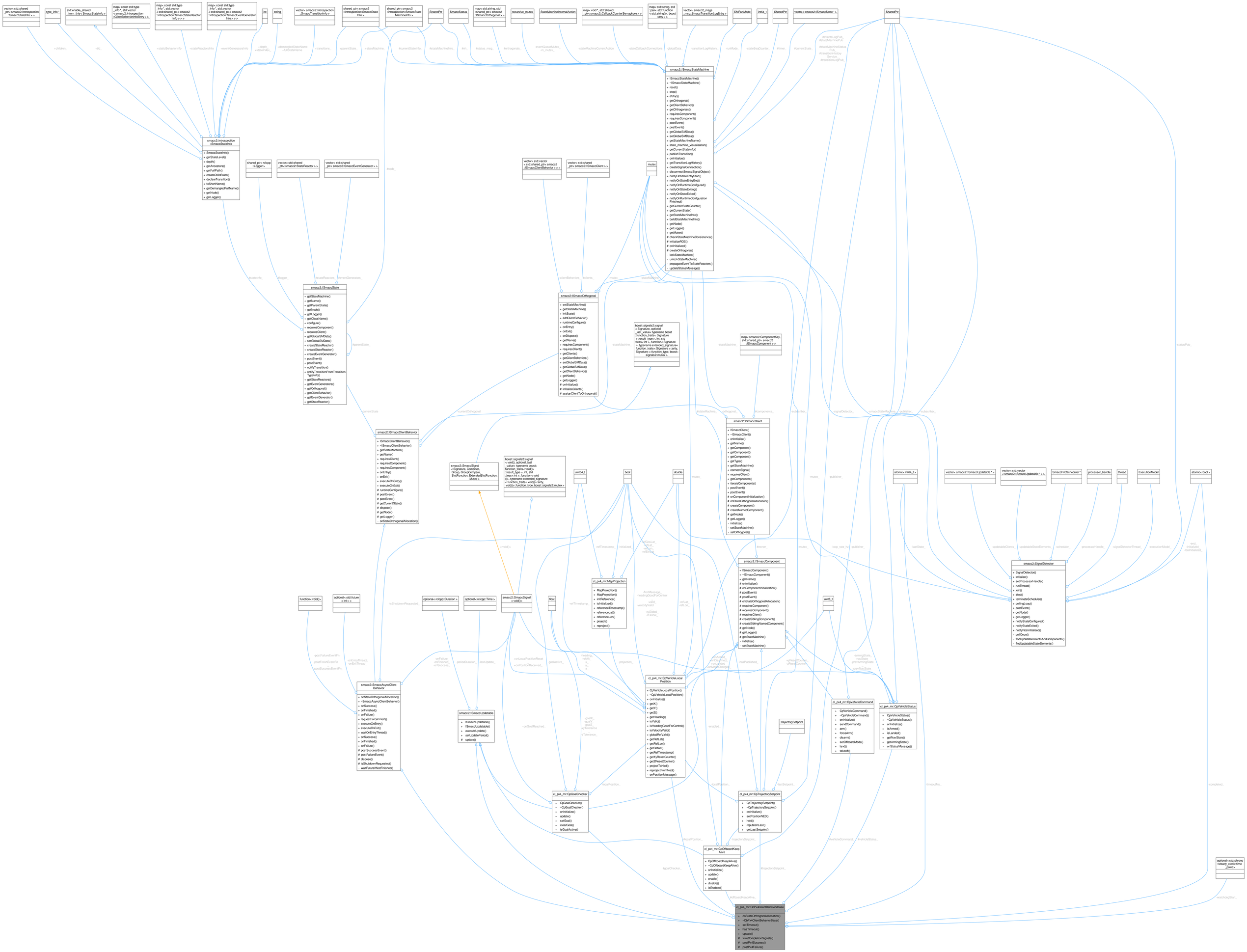
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::unable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
int
string
vector< smacc2::introspection::SmaccTransitionInfo >
shared_ptr< smacc2::introspection::SmaccStateInfo >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
SharedPtr
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
SharedPtr
+children_
+id_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
-depth_
+stateIndex_
+demangledStateName
+fullStateName
+transitions_
+parentState_
+stateMachine_
#currentStateInfo_
#stateMachineInfo_
#rtti_
#status_msg_
#orthogonals_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# oninitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
SharedPtr
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
#statusPub_
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
#node_
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
#currentState
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
vector< std::shared_ptr< smacc2::ISmaccClient > >
mutex
#clientBehaviors_
#clients_
#mutex_
#stateMachine_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
#stateMachine_
#orthogonal_
#components_
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#owner_
#mutex_
#loop_rate_hz
#publisher_
#lastState_
#updatableClients_
#updatableStateElements_
#scheduler_
#processorHandle_
#signalDetectorThread_
#executionModel_
#end_initialized_
#isInitialized_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void (), typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
uint64_t
bool
double
#mutex_
#mutex_
publisher_
atomic< int64_t >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
atomic< bool >
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
#signalDetector_
#smaccStateMachine_
#subscriber_
#subscriber_
#subscriber_
< void()>
#refTimestamp_
#initialized_
#refCostLat_ #refLat_ #refLon_ #refSet_
#owner_
#mutex_
#loop_rate_hz
#publisher_
#lastState_
#updatableClients_
#updatableStateElements_
#scheduler_
#processorHandle_
#signalDetectorThread_
#executionModel_
#end_initialized_
#isInitialized_
#smaccMs_
#completed_
#watchdogStart_
function< void()>
optional< std::future< int > >
optional< rclcpp::Duration >
optional< rclcpp::Time >
smacc2::SmaccSignal< void()>
float
uint8_t
#isShutdownRequested_
#refTimestamp_
#firstMessage_ #headingGoodForControl
#valid_ #velocityValid
#refLat_ #refLon_
#gvGlobal_ #rGlobal_
#postFailureEventFn
#postFinishEventFn
#postSuccessEventFn_
#onEntryThread_
#onExitThread_
#onFailure_ #onFinished_ #onSuccess_
#periodDuration_
#lastUpdate_
#onLocalPositionReset
#onPositionReceived_
#goalActive_
#projection_
#hasPublished_
#gvResetCounter_ #rResetCounter_
#gvArming State_ #prevArmingState_
#gvNavState_ #prevNavState_
smacc2::SmaccAsyncClientBehavior
+ orStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
cl_p4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
cl_p4_mr::ClpVehicleLocal Position
+ ClpVehicleLocalPosition()
+ ~ClpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
cl_p4_mr::ClpVehicleCommand
+ ClpVehicleCommand()
+ ~ClpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
cl_p4_mr::ClpVehicleStatus
+ ClpVehicleStatus()
+ ~ClpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
#stateMachine_
#orthogonal_
#components_
#subscriber_
#signalDetector_
#smaccStateMachine_
#publisher_
#subscriber_
#subscriber_
#statusPub_
cl_p4_mr::ClpGoalChecker
+ ClpGoalChecker()
+ ~ClpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
cl_p4_mr::ClpTrajectorySetpoint
+ ClpTrajectorySetpoint()
+ ~ClpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
TrajectorySetpoint
cl_p4_mr::ClpOffboardKeep Alive
+ ClpOffboardKeepAlive()
+ ~ClpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
cl_p4_mr::ClpPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~ClpPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
optional< std::chrono::steady_clock::time_point >
#currentOrthogonal
#goalX_ #goalY_ #goalZ_ #xyTolerance_ #zTolerance_
#onGoalReached_
#localPosition_
#localPosition_
#lastSetpoint_
#enabled_
#offboardKeepAlive_
#goalChecker_
#trajectorySetpoint_
#vehicleCommand_
#vehicleStatus_
#watchdogStart_